

```
In [16]: import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
```

```
In [2]: df = pd.read_csv("C:/Users/pratiksha/Downloads/Academic-Performance-Dataset.csv")
```

```
In [3]: df.head()
```

```
Out[3]:
```

	Rollno	Name	Gender	Branch	Attendance	Phy_marks	Che_marks	EM1_marks	PPS_marks	SME_m
0	1	Mohammed	M	Comp	72.0	62.0	98.0	63.0	89.0	
1	2	Reyansh	M	IT	58.0	62.0	83.0	83.0	88.0	
2	3	Aarav	M	IT	57.0	-20.0	100.0	NaN	56.0	
3	4	Atharv	M	IT	60.0	89.0	83.0	70.0	33.0	
4	5	Vivaan	M	Comp	85.0	90.0	NaN	78.0	23.0	



```
In [4]: df.shape
```

```
Out[4]: (20, 12)
```

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In [5]: df.dtypes
```

```
Out[5]: Rollno      int64
Name      object
Gender     object
Branch     object
Attendance float64
Phy_marks  float64
Che_marks  float64
EM1_marks  float64
PPS_marks  float64
SME_marks  float64
Total Marks int64
Percentage float64
dtype: object
```

```
In [6]: df.isna().sum()
```

```
Out[6]: Rollno      0
Name      2
Gender     0
Branch     0
Attendance 0
Phy_marks  1
Che_marks  3
EM1_marks  2
PPS_marks  1
SME_marks  0
Total Marks 0
Percentage 0
dtype: int64
```

```
In [7]: cols_with_na = []
for col in df.columns:
    if df[col].isna().any():
        cols_with_na.append(col)

cols_with_na
```

```
Out[7]: ['Name', 'Phy_marks', 'Che_marks', 'EM1_marks', 'PPS_marks']
```

```
In [15]: for col in cols_with_na:
col_dt = df[col].dtypes
if (col_dt == 'int64' or col_dt == 'float64'):
    outliers = (df[col] < 0) | (100 < df[col])
    df.loc[outliers, col] = np.nan
    df[col] = df[col].fillna(df[col].mean())
else:
    df[col] = df[col].ffill()
df.head()
```

Out[15]:

	Rollno	Name	Gender	Branch	Attendance	Phy_marks	Che_marks	EM1_marks	PPS_marks	SME_r
0	1	Mohammed	M	Comp	72.0	62.000000	98.000000	63.000000	89.0	
1	2	Reyansh	M	IT	58.0	62.000000	83.000000	83.000000	88.0	
2	3	Aarav	M	IT	57.0	74.058824	100.000000	83.444444	56.0	
3	4	Atharv	M	IT	60.0	89.000000	83.000000	70.000000	33.0	
4	5	Vivaan	M	Comp	85.0	90.000000	80.764706	78.000000	23.0	

```
In [9]: df.isna().sum()
```

```
Out[9]: Rollno      0
Name            0
Gender          0
Branch          0
Attendance      0
Phy_marks       0
Che_marks       0
EM1_marks       0
PPS_marks       0
SME_marks       0
Total Marks     0
Percentage      0
dtype: int64
```

```
In [19]: df_clean = df[(df['Che_marks'] >= Lower_limit) &
(df['Che_marks'] <= Upper_limit)].copy()
```

```
In [20]: scaler = MinMaxScaler()

df_clean['Percentage_scaled'] = scaler.fit_transform(df_clean[['Percentage']])

print(df_clean[['Percentage', 'Percentage_scaled']].head())
```

	Percentage	Percentage_scaled
0	73.6	0.712551
1	70.0	0.639676
2	38.4	0.000000
3	59.6	0.429150
4	49.4	0.222672

```
In [10]: Q1 = df['Che_marks'].quantile(0.25)
Q3 = df['Che_marks'].quantile(0.75)
IQR = Q3 - Q1

Lower_limit = Q1 - 1.5 * IQR
Upper_limit = Q3 + 1.5 * IQR

print(f'Q1 = {Q1}, Q3 = {Q3}, IQR = {IQR}, Lower_limit = {Lower_limit}, Upper_limit = {Upper_limit}')
```

Q1 = 76.0, Q3 = 86.25, IQR = 10.25, Lower_limit = 60.625, Upper_limit = 101.625

```
In [11]: df[(df['Che_marks'] < Lower_limit) | (df['Che_marks'] > Upper_limit)]
```

Out[11]:

	Rollno	Name	Gender	Branch	Attendance	Phy_marks	Che_marks	EM1_marks	PPS_marks	SME_marl
7	8	Ishaan	M	ENTC	75.0	66.0	51.0	83.0	69.611111	76
14	15	Maryam	F	IT	64.0	87.0	60.0	90.0	65.000000	90

In [12]:

```
outliers = df[(df['Che_marks'] < Lower_limit) | (df['Che_marks'] > Upper_limit)]
print(outliers)
```

	Rollno	Name	Gender	Branch	Attendance	Phy_marks	Che_marks	EM1_marks	\
7	8	Ishaan	M	ENTC	75.0	66.0	51.0	83.0	
14	15	Maryam	F	IT	64.0	87.0	60.0	90.0	

	PPS_marks	SME_marks	Total Marks	Percentage
7	69.611111	76.0	192	38.4
14	65.000000	90.0	392	78.4

In [13]:

```
df_clean = df[(df['Che_marks'] >= Lower_limit) & (df['Che_marks'] <= Upper_limit)]
```

In [14]:

```
print(df_clean.shape)
```

(18, 12)

In []: